

Qatar-1b

R-2287

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120503 · created 2026-08-17T00:36:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456050.91977 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.011
Transit depth (Rp/Rs)^2	3.69 +/- 0.41 [%]
Orbital Inclination (inc)	82.99 +/- 0.53
Airmass coefficient 1 (a1)	1.227 +/- 0.017
Airmass coefficient 2 (a2)	-0.147 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	2.73
Optimal Annulus	10.14
Transit Duration (day)	0.0667 +/- 0.0036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.011 vs 0.1463 (deviation 4.15σ)
Duration fit vs published	0.0667 d vs 0.0692 d (deviation 0.69σ)
Mid-time O–C	-0.3 min
Depth SNR	17.5
Residual scatter	1.36 %

Light curve

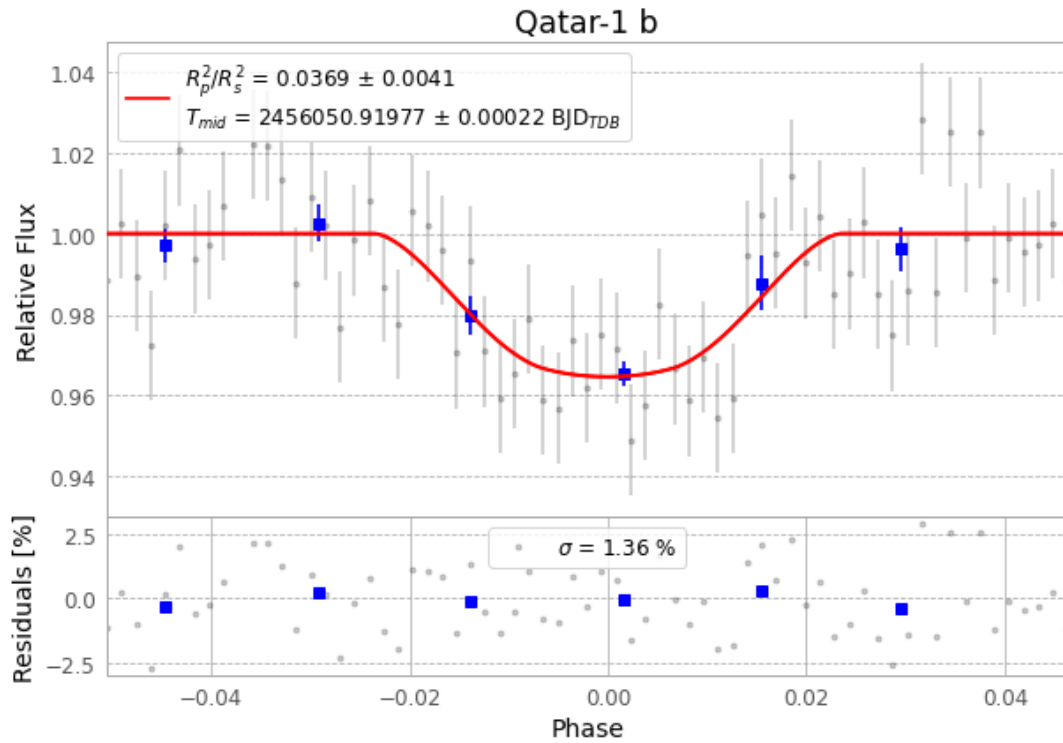


Figure 1 — FinalLightCurve_Qatar-1 b_2012-05-03.png (SHA-256 9691ec92dd348615...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [426, 181], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2b0bf1dc7c15ba8f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

67 FITS frames (44 MB), Qatar-1120503082112.FITS → Qatar-1120503113912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120503.log (d8ea08ae4961...), FinalLightCurve_Qatar-1 b_2012-05-03.png (9691ec92dd34...), FinalLightCurve_Qatar-1 b_2012-05-03.csv (645e4b4b331f...)

Compute resources

Wall time 127.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:36Z.